



Yakeen Plus (2025)

Practice Test-19

DURATION : 200 Minutes

DATE : 30/12/2024

M. MARKS : 720

Topics Covered

Physics :	Gravitation, Properties of Solids and Liquids, Thermodynamics, Kinetic Theory of Gases Oscillation and Waves
Chemistry :	Redox Reaction, Electrochemistry, Hydrocarbon, Haloalkanes and Haloarenes, Coordination Compound
Biology :	(Botany): Morphology of Flowering Plants, Anatomy of Flowering Plants, Respiration in Plants Photosynthesis in Higher Plants, Plant Growth and Development (Zoology): Body Fluids and Circulation, Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are **four Section** in the Question Paper, **Section I, II, III & IV** consisting of Section-I (Physics from question number **1 to 50**), Section-II (Chemistry: from question number **51 to 100**), Section-III (Botany: from question number **101 to 150**) & Section IV (Zoology: from question number **151 to 200**), having 50 Questions in each Subject and each subject is divided in two Section, Section A consisting 35 questions (all questions are compulsory) and Section B consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

Answer Key

Q1 (1)
Q2 (1)
Q3 (2)
Q4 (3)
Q5 (1)
Q6 (4)
Q7 (3)
Q8 (1)
Q9 (3)
Q10 (4)
Q11 (2)
Q12 (2)
Q13 (3)
Q14 (3)
Q15 (1)
Q16 (3)
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Q18 (1)
Q19 (4)
Q20 (3)
Q21 (2)
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Q63 (4)
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Q195 (4)

Q196 (1)

Q197 (4)

Q198 (2)

Q199 (1)

Q200 (3)



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Hints & Solutions

Q1 Text Solution:

(1)

Average kinetic energy of a gas molecule

$$k \cdot E_{avg} = \frac{f}{2} kT \left\{ f = \text{degree of freedom} \right\}$$

A. $T \downarrow \Rightarrow K. E_{avg} \downarrow$

B. $T = \text{constant} \Rightarrow K. E_{avg} = \text{constant}$

C. Not necessary

D. Pressure is constant given

E. Not necessary

Q2 Text Solution:

(1)

$$h = \frac{2T \cos \theta}{\rho r g}$$

here, r is radius of tube.

Q3 Text Solution:

(2)

$$E = KE + PE$$

$$E = \frac{1}{2} m \omega^2 (A^2 - x^2) + \frac{1}{2} m \omega^2 x^2$$

$$\text{also, } \omega = \sqrt{\frac{k}{m}}$$

$$\text{If } m \text{ is doubled, } \omega' = \frac{\omega}{\sqrt{2}}$$

Here, value of $m \omega^2$ remains constant

Q4 Text Solution:

(3)

$$\text{We know, } F_v = 6\pi\eta r v$$

$$\therefore \text{ and } v \propto r^2 \frac{F_1}{F_2} = \frac{r_1^3}{r_2^3}$$

$$= \frac{V}{2V}$$

$$\text{or } F_2 = 2F_1$$

Q5 Text Solution:

(1)

CH_4 is non-linear and polyatomic

Q6 Text Solution:

(4)

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$T^2 = \frac{4\pi^2 l}{g'} \quad \dots (1)$$

$$g' = g \left(\frac{R}{R+2R} \right)^2$$

$$g' = \frac{g}{9} \quad \dots (2)$$

$$T^2 = \frac{4\pi^2 l}{g/9}$$

$$\frac{2^2 \times g}{9 \times 4\pi^2} = 1$$

$$1 = \frac{1}{9} m$$

Q7 Text Solution:

(3)

$$\frac{\text{Strain}}{\text{Stress}} = \frac{1}{Y}$$

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Q8 Text Solution:

(1)

$$\frac{g'}{g} = \left(\frac{R}{r} \right)^2$$

$$g' = g \left(\frac{\frac{R}{5}}{\frac{4}{5}R} \right)^2$$

$$= g \left(\frac{4R}{5R} \right)^2$$

$$g' = \frac{16}{25} g$$

$$\Delta g = g - g'$$

$$= g - \frac{16}{25} g$$

$$= \frac{9}{25} g$$

$$\frac{\Delta g}{g} \times 100\% = \frac{9}{25} \times 100 = 36\%$$

Q9 Text Solution:

(3)

$$\text{Using } PV = nRT = \frac{m}{M} RT$$

$$\therefore \frac{m_1}{m_2} = \frac{P_1 V_1}{P_2 V_2} = \frac{\left(\frac{4T}{R_1} \right) \times \frac{4}{3} \pi R_1^3}{\left(\frac{4T}{R_2} \right) \times \frac{4}{3} \pi R_2^3}$$

$$= \frac{R_1^2}{R_2^2}$$

Q10 Text Solution:

(4)



irreversible adiabatic $\rightarrow$ Because Energy can not be restored.

Q11 Text Solution:

(2)

$$v = \omega \sqrt{A^2 - x^2}$$

$$v^2 = \omega^2 A^2 - \omega^2 x^2$$

$$v^2 + \omega^2 x^2 = \omega^2 A^2$$

$$\frac{v^2}{(\omega A)^2} + \frac{x^2}{A^2} = 1 \rightarrow \text{Elliptical}$$

Q12 Text Solution:

(2)

$$n = 7$$

$$\Delta T = 40 \text{ K}$$

$$\Delta U = \frac{n f R \Delta T}{2}$$

$$= \frac{7 \times 3 \times 8.3 \times 40}{2} = 21 \times 166 = 3486 \text{ J}$$

Q13 Text Solution:

(3)

$$V_1 = V$$

So

$$V_2 = \frac{V}{8}$$

$$P_1 V_1^r = P_2 V_2^r$$

$$P_1 V^{5/3} = P_2 \left(\frac{V}{8}\right)^{5/3}$$

$$\boxed{P_2 = 32 P_1}$$

Q14 Text Solution:

(3)

$$a_1 v_1 = a_2 v_2 \Rightarrow \frac{v_2}{v_1} = \frac{a_1}{a_2} = \left(\frac{r_1}{r_2}\right)^2$$

$$\Rightarrow v_2 = 3 \times (2)^2 = 12 \text{ m/s}$$

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Q15 Text Solution:

(1)

$$g' = g \left(1 - \frac{2h}{R}\right)$$

$$g' = g \left(1 - \frac{2 \times 32}{6400}\right) = \frac{g(99)}{100}$$

$$\frac{mg - mg'}{mg} \times 100 = \frac{g - g'}{g} \times 100 = \frac{g - \frac{99g}{100}}{g} \times 100$$

$$= \frac{1}{100} \times 100 = 1\%$$

Q16 Text Solution:

(3)

A stream lined body has less resistance due to air.

[New NCERT Class 11th Page No. 189]

Q17 Text Solution:

(1)

$$C_v = \frac{nR}{2}$$

$$C_p = \frac{nR}{2} + R = \frac{n+2}{2} R$$

$$\frac{C_v}{C_p} = \frac{n}{n+2}$$

Q18 Text Solution:

(1)

$$v_{max} = v_{wave}$$

$$\Rightarrow A\omega = \frac{\omega}{k}$$

$$\Rightarrow A = \frac{1}{k}$$

$$\Rightarrow 2 = \frac{\lambda}{2\pi}$$

$$\Rightarrow \lambda = 4\pi$$

Q19 Text Solution:

(4)

$$g = \frac{GM}{R^2}$$

$$M = \text{constant } g \propto \frac{1}{R^2}$$

$$100 \frac{\Delta g}{g} = -2 \frac{\Delta R}{R} 100$$

$$\% \text{ change} = -2 (-2)$$

$$\% \text{ change in } g = 4\%$$

$$\text{increase by } 4\%$$

Q20 Text Solution:

(3)

$$F = \gamma A \frac{\Delta l - l}{l}$$

$$= 2 \times 10^{11} \times 10^{-4} \left(\frac{2l-1}{1}\right)$$

$$= 2 \times 10^7 \text{ N}$$

Q21 Text Solution:

(2)

$$[P_{avg} = 0] \text{ (due to random motion)}$$

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$T_{new} = 2T$$

$$M_{new} = \frac{M}{2}$$

$$\frac{v_{new}}{v} = \frac{\sqrt{\frac{2T}{M/2}}}{\sqrt{\frac{T}{M}}}$$

$$v_{new} = 2v$$

Q22 Text Solution:

(3)



$$y = 0.5 \sin \left(\frac{2\pi}{\lambda} 400t - \frac{2\pi}{\lambda} x \right)$$

$$\omega = \frac{2\pi}{\lambda} 400$$

$$K = \frac{2\pi}{\lambda}$$

$$v = \frac{\omega}{k} \quad [v = 400 \text{ m/s}]$$

Q23 Text Solution:

(4)

$$P_{\text{in}} = P_0 + \frac{4T}{R_1}$$

$$1.01 = 1 + \frac{4T}{R_1}$$

$$0.01 = \frac{4T}{R_1}$$

$$0.02 = \frac{4T}{R_2}$$

$$\frac{1}{2} = \frac{R_2}{R_1}$$

$$\frac{V_1}{V_2} = \frac{8}{1}$$

Q24 Text Solution:

(1)

Liquid surface will be straight line for liquid moving with constant acceleration.

$$\text{Hence slope } \tan \theta = \frac{a}{g}$$

[New NCERT Class 11th Page No. 186]

Q25 Text Solution:

(1)

$$\Delta l = l \propto \Delta t$$

$$\alpha = \frac{\Delta l}{l \Delta t} = \frac{0.02}{100 \times 10} = 2 \times 10^{-5}$$

$$v = 3\alpha = 6 \times 10^{-5}$$

$$\begin{aligned} \text{Now, } \frac{\Delta v}{v} \times 100 &= \gamma \cdot \Delta t \cdot 100 = 6 \times 10^{-5} \\ &\times 10 \times 100 \\ &= 6 \times 10^{-2} = 0.06 \end{aligned}$$

Q26 Text Solution:

(4)

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\frac{V_{\text{He}}}{V_{\text{O}_2}} = \sqrt{\frac{M_{\text{O}_2}}{M_{\text{He}}}} = \sqrt{\frac{32}{4}} = \sqrt{8} = 2\sqrt{2}$$

Q27 Text Solution:

(2)

Horizontal range will be maximum when $h = \frac{H}{2} = \frac{90}{2} = 45 \text{ cm}$ i.e. hole 3.

[New NCERT Class 11th Page No. 189-190]

Q28 Text Solution:

(4)

$$\frac{F}{A_1} = \frac{\text{weight of car}}{A_2}$$

$$F = mg \left(\frac{A_1}{A_2} \right) = 20000 \left(\frac{1 \text{ cm}^2}{1000 \text{ cm}^2} \right) = 20 \text{ N}$$

[New NCERT Class 11th Page No. 192]

Q29 Text Solution:

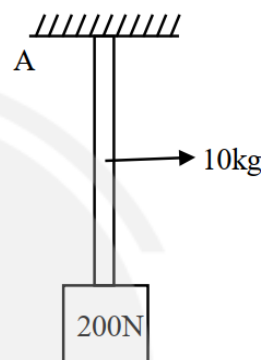
(3)

Property of viscous flow. The layer in away most from the surface of pipe will have maximum velocity.

[New NCERT Class 11th Page No. 191]

Q30 Text Solution:

(1)



$$T_A = 10 \text{ g} + 200$$

$$T_A = 300 \text{ N}$$

Q31 Text Solution:

(2)

$$P_{\text{ex}} = P_{\text{in}} - P_{\text{out}}$$

$$P_{\text{in}} = P_{\text{out}} + P_{\text{axe}}$$

$$P_{\text{in}} = P_{\text{out}} + \frac{4s}{R}$$

Q32 Text Solution:

(2)

$\therefore$ For adiabatic process

$$Pv^\gamma = k$$

$$P_1 v_1^{3/2} = P_2 \times (4)^{3/2}$$

$$P_1 \times (5)^{3/2} = P_2 \times (4)^{3/2}$$

$$\frac{P_1}{P_2} = \left(\frac{4}{5} \right)^{3/2} = \frac{(2)^3}{(125)^{1/2}} = \frac{8}{5\sqrt{5}}$$

Q33 Text Solution:

(3)

$$\text{Excess pressure in soap bubble} = \frac{4S}{R}$$

$$\text{Given } P_1 = 3P_2$$



$$\therefore P_1 = \frac{4S}{R_1} \& P_2 = \frac{4S}{R_2}$$

$$3P_2 = \frac{4S}{R_1}; P_2 = \frac{4S}{R_2}$$

$$\therefore R_1 = \frac{4S}{3P_2}; R_2 = \frac{4S}{P_2}$$

$$\frac{R_1}{R_2} = \frac{4S}{3P_2} \times \frac{P_2}{4S} = \frac{1}{3}$$

$\therefore$ Ratio between the volume :

$$\frac{V_1}{V_2} = \frac{\frac{4}{3}\pi R_1^3}{\frac{4}{3}\pi R_2^3} = \frac{1}{27}$$

Q34 Text Solution:

(3)

$$\frac{W_1}{W_2} = \frac{8\pi r_1^2 T}{8\pi r_2^2 T} = \frac{r_1^2}{r_2^2}$$

$$\text{As } V = \frac{4}{3}\pi r^3, \therefore \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{V}{2V}\right)^{\frac{2}{3}}$$

Q35 Text Solution:

(3)

If the liquid is incompressible then mass of liquid entering through left end, should be equal to mass of liquid coming out from the right end.

$$\therefore M = m_1 + m_2 \Rightarrow Av_1 = Av_2 + 1.5A$$

$\cdot v$

$$\Rightarrow A \times 3 = A \times 1.5 + 1.5A \cdot v$$

$$\Rightarrow v = 1 \text{ m/s}$$

[New NCERT Class 11th Page No. 191]

Q36 Text Solution:

(1)

$$T = 2\pi\sqrt{\frac{3m}{k_{eq}}} \left\{ k_{eq} = 2k \right\}$$

$$T_1 = 1\pi\sqrt{\frac{3m}{2k}}$$

$$T_2 = 2\pi\sqrt{\frac{m}{3k}}$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{9}{2}} \Rightarrow \frac{T_1}{T_2} = \frac{3}{\sqrt{2}}$$

Q37 Text Solution:

(1)

$$Q = \frac{mgh}{2J} = \frac{(\pi r^2 h)\rho h}{2J} = \frac{\pi r^2 h^2 \rho g}{2J}$$

Q38 Text Solution:

(1)

$$P_{av} \times (lh) = Tl$$

$$\text{or } \frac{\rho gh}{2} \times lh = Tl$$

$$\therefore h = \sqrt{\frac{2T}{\rho g}}$$

Q39 Text Solution:

(3)

$$U = n_1 C_{v_1} T + n_2 C_{v_2} T$$

$$= 3 \times \frac{5}{2} RT + 5 \times \frac{3}{2} RT$$

$$= \frac{30}{2} RT = 15RT$$

Q40 Text Solution:

(1)

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

given, $\mu_x = \mu_z$ & $L_x = L_z$ as identical

$$\therefore f \propto \sqrt{T}$$

$$\Rightarrow \frac{T_x}{T_z} = \frac{f_x^2}{f_z^2} = \left(\frac{450}{300}\right)^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$$

$$\frac{T_x}{T_y} = 2.25$$

Q41 Text Solution:

(4)

at constant velocity

$$F_B + F_v = mg$$

$$F_v = mg - F_B$$

$$= mg \left(1 - \frac{\rho_0}{\rho}\right)$$

$$F_B = \rho_0 \cdot \left(\frac{m}{\rho}\right) \cdot g$$

Q42 Text Solution:

(1)

$$\lambda = \frac{1}{\sqrt{2} \left(\frac{N}{V}\right) \pi d^2}$$

$\lambda \rightarrow$ Mean free path

$N \rightarrow$ No. of molecules

$v =$ volume of container

$d =$ diameter of molecule

$\therefore N$ and V are constant

$\therefore$ Mean free path remains unchanged.

Now, If $T \uparrow$ no. of collisions increases.

Q43 Text Solution:

(1)

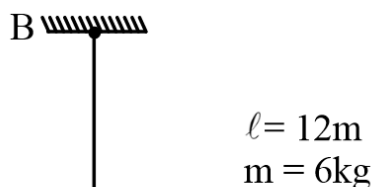
Hooke's law $\rightarrow$ stress $\propto$ strain

[New NCERT Class 11th Page No. 172]

Q44 Text Solution:

(1)





$$\mu = \frac{6}{12} = \frac{1}{2} \text{ kg/m}$$

$$A \rightarrow V = \sqrt{\frac{T}{\mu}} = f\lambda$$

$$B \rightarrow \sqrt{\frac{T'}{\mu}} = f\lambda'$$

$$\sqrt{\frac{T}{T'}} = \frac{\lambda}{\lambda'}$$

$$\sqrt{\frac{20}{80}} = \frac{6}{\lambda'} = \lambda' = 12$$

Q45 Text Solution:

(2)

$$U = \frac{f}{2} nRT$$

$$U = \frac{6}{2} nRT$$

$$U = 3nRT \quad n = 1$$

$$U = 3RT$$

Q46 Text Solution:

(1)

$$\frac{F_1}{F_2} = \frac{T \times \pi \frac{(a+b)}{2}}{T \times 2b} = \frac{\pi(a+b)}{4b}$$

Q47 Text Solution:

(3)

$$T_1 = 4\text{s} \quad T_2 = 6\text{s}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\text{So, } \frac{T_1}{T_2} = \sqrt{\frac{g_2}{g_1}} \quad \dots (1)$$

$$g_1 = \frac{GM}{R^2}$$

$$g_2 = \frac{GM}{(R+h)^2}$$

$$\text{So, } \frac{g_2}{g_1} = \frac{R^2}{(R+h)^2}$$

From equation (1)

$$\frac{T_1}{T_2} = \sqrt{\frac{R^2}{(R+h)^2}}$$

$$\Rightarrow \frac{4}{6} = \frac{R}{R+h}$$

$$4R + 4h = 6R$$

$$4h = 2R$$

$$h = R/2$$

$$= \frac{6400}{2} = 3200 \text{ km}$$

Q48 Text Solution:

(1)

$$F = P \times A$$

$$= \rho gh \times A$$

$$= 900 \times 10 \times 0.4 \times 10^{-3} = 3.6 \text{ N}$$

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Q49 Text Solution:

(4)

$$1.5 = (2n_1 + 1) \lambda/2 \quad \dots(1)$$

$$5 = n_2 \lambda \quad \dots(2)$$

n_1 & n_2 are integer

$$n_1 = 1, n_2 = 5$$

$$n_1 = 2, n_2 \text{ is not integer}$$

$$n_1 = 3, n_2 \text{ is not integer}$$

$$n_1 = 4, n_2 = 15, \quad \lambda = 1/3$$

Q50 Text Solution:

(2)

Terminal velocity

$$v = \frac{2}{9} r^2 \frac{(\rho - \sigma)}{\eta} g$$

$$= \frac{2}{9} \times 10^{-8} \frac{(10^5 - 10^3) \times 9.8}{9.8 \times 10^{-6}}$$

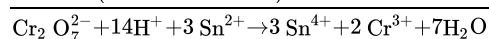
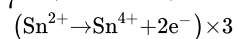
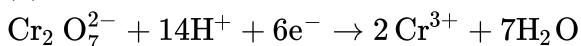
$$= \frac{2}{9} \times 10^{-8} \frac{99 \times 10 \times 9.8}{9.8 \times 10^{-6}}$$

$$= 22 \times 10 = 220 \text{ m/s}$$

$$\text{Now } h = \frac{v^2}{2g} = \frac{48400}{2 \times 9.8} \approx 2518 \text{ m}$$

Q51 Text Solution:

(3)



It is clear from this equation that 3 moles of Sn^{2+} reduce 1 mol of $\text{Cr}_2 \text{O}_7^{2-}$, hence 1 mol of Sn^{2+} will reduce $\frac{1}{3}$ moles of $\text{Cr}_2 \text{O}_7^{2-}$.

Q52 Text Solution:



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(3)

When acetylene and hydrogen are passed over heated Lindlar's catalyst, partial hydrogenation takes place and ethylene is formed.

Q53 Text Solution:

(1)

Applying $w = Z \times l \times t$

1 millimole of Al = 27×10^{-3} g

$$Z = \frac{\text{Eq. wt.}}{96500} = \frac{27/3}{96500} = \frac{9}{96500}$$

$$t = 27 \times 10^{-3} \times \frac{96500}{9} \times \frac{1}{9.65} = 30 \text{ seconds}$$

Q54 Text Solution:

(3)

In $K_4[Fe(CN)_6]$, the central metal Fe is surrounded by 6 CN^- ligands.

Hence, coordination no. of Fe is 6.

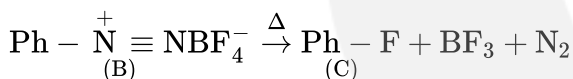
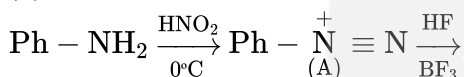
Oxidation state of Fe = $4 + x + (-6) = 0$

$$x - 2 = 0$$

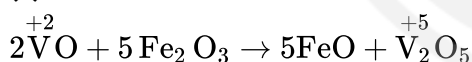
$$x = +2$$

Q55 Text Solution:

(4)

**Q56 Text Solution:**

(3)



$$n = 6$$

$$\therefore \text{Equivalent Weight} = \frac{M}{6}$$

Q57 Text Solution:

(2)

Electrophilic addition of Br_2 to alkene involves anti addition of Br_2 across the double bond.

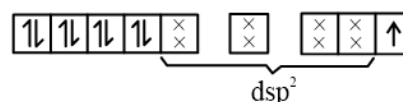
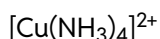
Q58 Text Solution:

(3)

$$Z = \frac{E}{96,500} = \frac{\text{At. wt.}}{\text{Valency}} \times \frac{1}{96500}$$

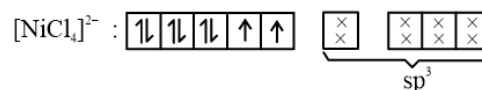
Q59 Text Solution:

(3)



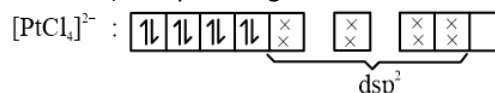
No. of unpaired electrons = 1

So, $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is paramagnetic.



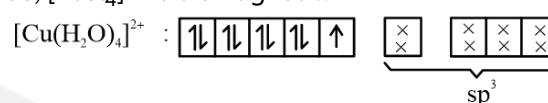
No. of unpaired electrons = 2

So, $[\text{NiCl}_4]^{2-}$ is paramagnetic.



No. of unpaired electrons = 0

So, $[\text{PtCl}_4]^{2-}$ is diamagnetic.



No. of unpaired electrons = 1

So, $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ is paramagnetic.

Q60 Text Solution:

(3)

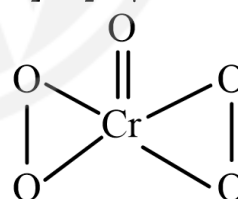
As the standard reduction potential drops, reducing power increases

So, $Z (-3.0 \text{ V}) > X (-1.2 \text{ V}) > Y (+0.5 \text{ V})$

Q61 Text Solution:

(3)

$$\text{K}_2\text{Cr}_2\text{O}_7 : 2 + 2x + 7(-2) = 0 \Rightarrow x = +6$$



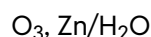
Peroxo oxygen has -1 oxidation state.

Let Oxidation state of Cr be 'x'

$$\text{CrO}_5 : x + 4(-1) + 1(-2) = 0 \Rightarrow x = +6$$

Q62 Text Solution:

(2)

**Q63 Text Solution:**

(4)

Primary cells are those cells, in which the reaction occurs only once and after use over a period of time, it becomes dead and cannot be reused again. e.g., Leclanche cell, used in transistors and



clocks and mercury cell suitable for low current devices. Lead storage battery and Ni-Cd cell are secondary cells.

Q64 Text Solution:

(1)

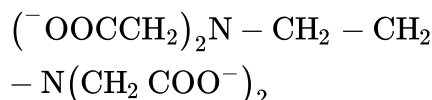
Small ions are solvated best in protic solvents.

Q65 Text Solution:

(3)

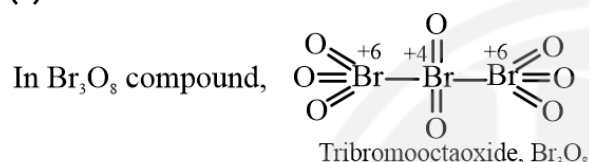
The structural formula of EDTA

(Ethylenediaminetetraacetate ion) is



Q66 Text Solution:

(4)

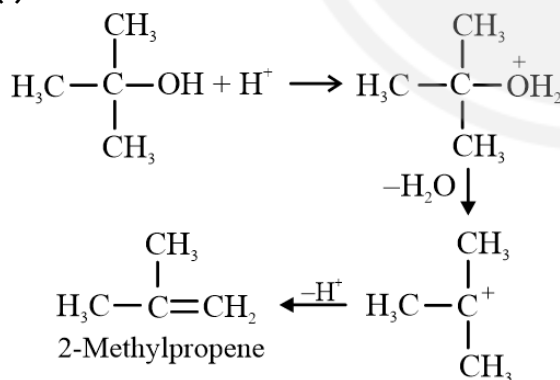


each of the two terminal bromine atoms are present in +6 oxidation states and the middle bromine is present in +4 oxidation state.

Hence, the overall oxidation number of bromine is 16/3.

Q67 Text Solution:

(1)



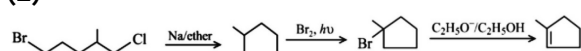
Q68 Text Solution:

(2)

In lead storage battery, lead (Pb) acts as anode and lead dioxide (PbO_2) act as cathode.

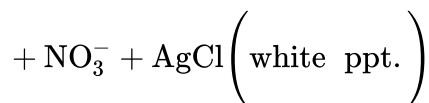
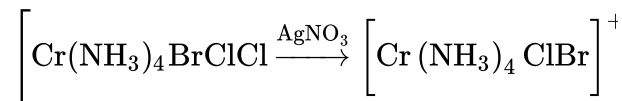
Q69 Text Solution:

(2)

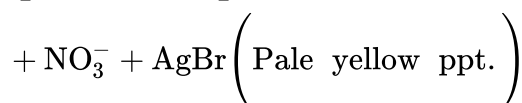
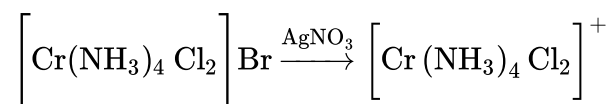


Q70 Text Solution:

(1)



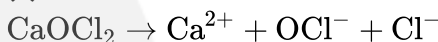
(X)



(Y)

Q71 Text Solution:

(3)



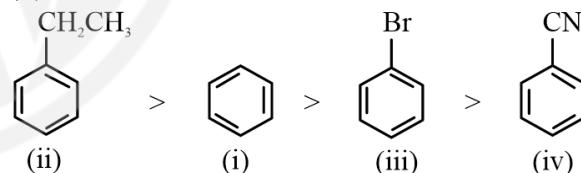
In OCl^- (hypochlorite ion oxidation state of Cl)

$$= -2 + x = -1 \Rightarrow x = +1$$

In Cl^- oxidation state of chlorine = -1

Q72 Text Solution:

(3)



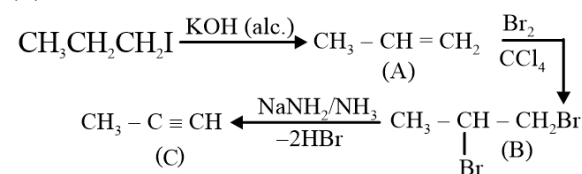
Q73 Text Solution:

(4)

Efficiency of fuel cell is very high.

Q74 Text Solution:

(3)



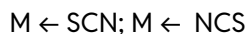
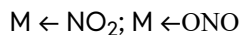
Q75 Text Solution:

(3)

There are certain ligands which have two or more donor atoms but in forming complexes only one

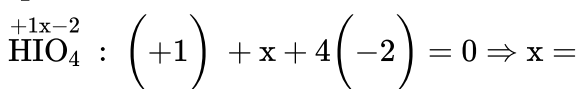
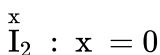
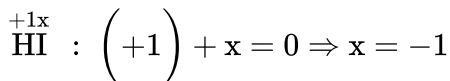
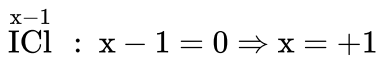


of the donor atoms is attached to metal ions, such ligands are called ambidentate ligands.



Q76 Text Solution:

(3)



+7

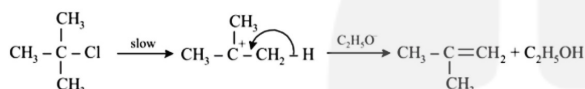
Thus decreasing order of oxidation number of iodine is



Q77 Text Solution:

(2)

It will undergo E1 reaction.



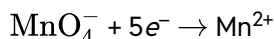
Q78 Text Solution:

(1)

Aromatic compounds are most stable followed by non-aromatic compounds and anti-aromatic compounds are least stable.

Q79 Text Solution:

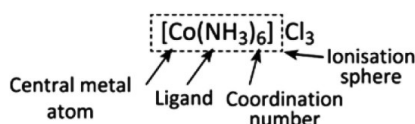
(3)



1 mol of MnO_4^- to Mn^{2+} requires 5 Faraday i.e., $5 \times 96500 \approx 4.83 \times 10^5 \text{ C}$

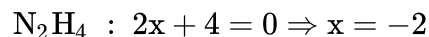
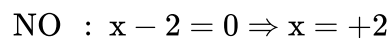
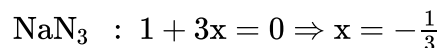
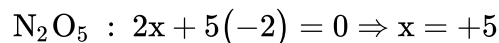
Q80 Text Solution:

(4)



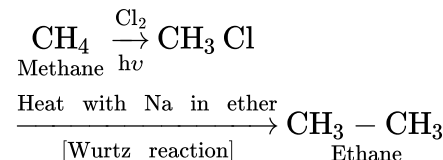
Q81 Text Solution:

(1)



Q82 Text Solution:

(3)



Q83 Text Solution:

(3)

Concentration of solution (c) = 0.02 M

$$\Lambda_m^c = 150 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}; \Lambda_m^o = 300 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

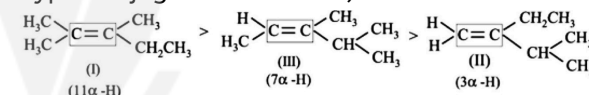
$$\alpha = \frac{\Lambda_m^c}{\Lambda_m^o} = \frac{150}{300} = 0.5$$

$$K_a = \frac{c\alpha^2}{1-\alpha} = \frac{0.02 \times (0.5)^2}{1-0.5} = 0.01 \text{ M}$$

Q84 Text Solution:

(2)

That alkene will be more stable on which number of α -hydrogens are maximum (more number of hyperconjugated structures).



Q85 Text Solution:

(1)

In bimolecular nucleophilic aromatic substitution reactions the order of reactivity of aryl halides is: $\text{ArF} > \text{ArCl} > \text{ArBr} > \text{ArI}$

Q86 Text Solution:

(1)

$[\text{Ni}(\text{NH}_3)_6]^{2+}$, sp^3d^2 (outer), octahedral, paramagnetic

$[\text{Zn}(\text{NH}_3)_6]^{2+}$, sp^3d^2 (outer), octahedral, diamagnetic

$[\text{Cr}(\text{NH}_3)_6]^{3+}$, d^2sp^3 (inner), octahedral, paramagnetic

$[\text{Co}(\text{NH}_3)_6]^{3+}$, d^2sp^3 (inner), octahedral, diamagnetic

Q87 Text Solution:

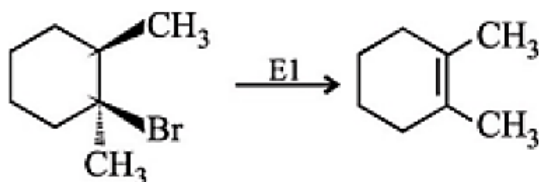
(3)



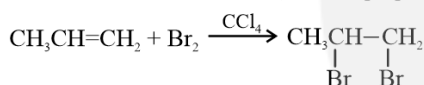
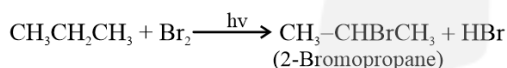
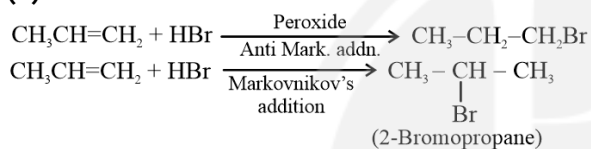
$$\begin{aligned}\text{Solubility} &= \frac{\kappa \times 1000}{\Lambda_m^\circ} \\ &= \frac{1.821 \times 10^{-5} \times 1000}{130.26} = 13.97 \times 10^{-5} \text{ mol L}^{-1} \\ &= 13.97 \times 10^{-5} \times 143.5 \\ &\quad (\text{AgCl} = 108 + 35.5 = 143.5) \\ &= 2.004 \times 10^{-2} \text{ g L}^{-1}\end{aligned}$$

Q88 Text Solution:

(2)

**Q89 Text Solution:**

(2)

**Q90 Text Solution:**

(1)

Cyclooctatetraene (tub shaped) is non-aromatic compound because of non-planarity and planar cyclooctatetraene is anti-aromatic due to 8π -electrons.

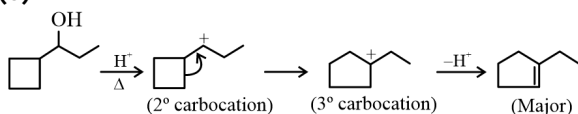
Q91 Text Solution:

(3)

When $-\text{NH}_3^+$, $-\text{COOR}$ and $-\text{CN}$ groups attached to benzene ring it will direct the incoming electrophile predominantly to the meta position.

Q92 Text Solution:

(3)

**Q93 Text Solution:**

(2)

Let the oxidation number of Cr in CrO_4^{2-} is x .

$$x - 8 = -2;$$

$$x = +6$$

Let the oxidation number of Cr in $\text{Cr}(\text{OH})_4^-$ is y .

$$y - 4 = -1 \text{ or } y = +3$$

Q94 Text Solution:

(1)

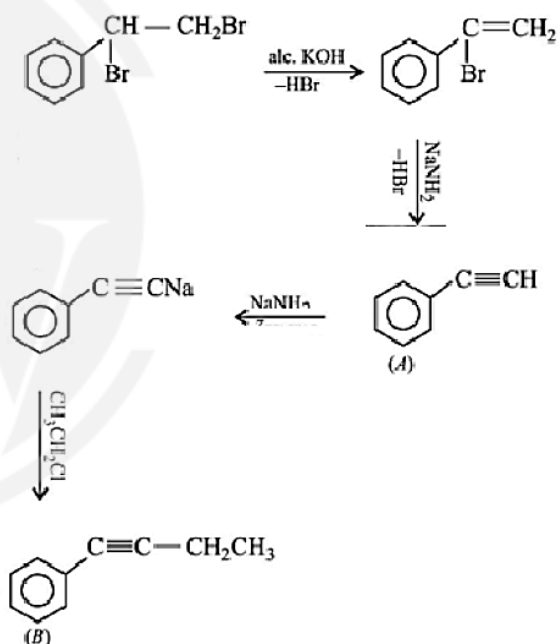
$$\begin{aligned}\Lambda_m \frac{\kappa}{C} &= \frac{4.95 \times 10^{-5} \text{ S cm}^{-1}}{0.001028 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}} \\ &= 48.15 \text{ S cm}^2 \text{ mol}^{-1}\end{aligned}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{48.15}{390.5} = 0.1233$$

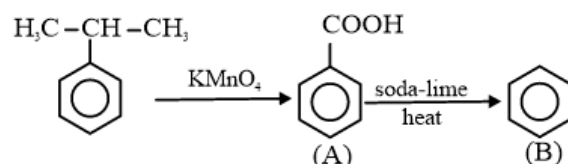
$$\begin{aligned}K_a &= \frac{C\alpha^2}{(1-\alpha)} = \frac{0.001028 \times (0.1233)^2}{1-0.1233} \\ &= 1.78 \times 10^{-5} \text{ mol L}^{-1}\end{aligned}$$

Q95 Text Solution:

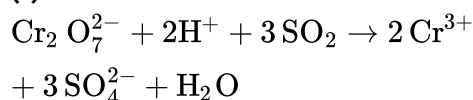
(2)

**Q96 Text Solution:**

(3)

**Q97 Text Solution:**

(1)



Q98 Text Solution:

(3)

Since the compound reacts with $[\text{Ag}(\text{NH}_3)_2]^+$, the compound must be terminal alkyne with group $\equiv \text{CH}$. On treatment with alkaline KMnO_4 compound gives $(\text{CH}_3)_3\text{C}-\text{COOH}$ which shows the presence of $(\text{CH}_3)_3\text{C}-\text{C}\equiv$ group. Thus, the compound should be $(\text{CH}_3)_3\text{C}-\text{C}\equiv\text{CH}$.

Q99 Text Solution:

(4)

According to Kohlrausch's law, Λ_m° for CH_3COOH
 $= \lambda_{\text{CH}_3\text{COO}^-}^\circ + \lambda_{\text{H}^+}^\circ$

$$\lambda^\circ \text{ for } \text{H}_2\text{SO}_4 = 2\lambda_{\text{H}^+}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ = x \text{ S cm}^2$$

$$\text{mol}^{-1} \dots\dots (\text{i})$$

$$\lambda^\circ \text{ for } \text{K}_2\text{SO}_4 = 2\lambda_{\text{K}^+}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ = y \text{ S cm}^2$$

$$\text{mol}^{-1} \dots\dots (\text{ii})$$

$$\lambda^\circ \text{ for } \text{CH}_3\text{COOK} = \lambda_{\text{CH}_3\text{COO}^-}^\circ + \lambda_{\text{K}^+}^\circ = z$$

$$\text{S cm}^2 \text{ mol}^{-1} \dots\dots (\text{iii})$$

on adding equation (1) and $2 \times (\text{iii})$ and subtracting (ii), we get

$$2\lambda_{\text{H}^+}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ + 2\lambda_{\text{CH}_3\text{COO}^-}^\circ + 2\lambda_{\text{K}^+}^\circ - 2\lambda_{\text{K}^+}^\circ - \lambda_{\text{SO}_4^{2-}}^\circ = x + 2z - y$$

$$\lambda_{\text{H}^+}^\circ + \lambda_{\text{CH}_3\text{COO}^-}^\circ = \frac{(x-y)}{2} + z$$

Q100 Text Solution:

(2)

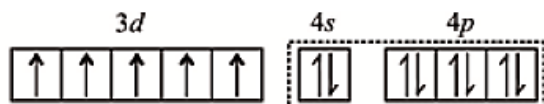
$\text{Cs}[\text{FeCl}_4]$, Caesium tetrachlorido ferrate (III)

Oxidation state = +3

Coordination number = 4

$\text{Fe} (26) : 3d^6 4s^2$

$\text{Fe}^{3+} : 3d^5 4s^0$



It is paramagnetic due to presence of 5 unpaired electrons.

Q101 Text Solution:

(3)

In alternate type of phyllotaxy, a single leaf arises at each node in alternate manner, as in china rose, mustard and sun flower plants. In opposite type, a pair of leaves arise at each node and lie opposite to each other as in *Calotropis* and guava plants. If more than two leaves arise at a node and form a whorl, it is called whorled, as in *Alstonia*.

[New NCERT Class 11th Page No. 61]

Q102 Text Solution:

(1)

The epidermal tissue system forms the outermost covering of the whole plant body and comprises epidermal cells, stomata and the epidermal appendages – the trichomes and hairs.

[New NCERT Class 11th Page No. 71, 72]

Q103 Text Solution:

(4)

RQ values $\rightarrow$

- Glucose - 1
- Fat - less than 1
- Tripalmitin - 0.7
- Protein - About 0.9

[New NCERT Class 11th Page No. 163, 164]

Q104 Text Solution:

(4)

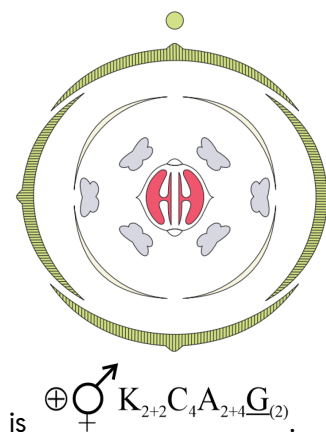
All green plants are autotrophs.

[New NCERT Class 11th Page No. 133]

Q105 Text Solution:

(2)

Floral formula of the given floral diagram.



Android App

iOS App

PW Website

[New NCERT Class 11th Page No. 67]**Q106 Text Solution:**

(3)

In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. When the bulliform cells in the leaves have absorbed water and are turgid, the leaf surface is exposed. When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss.

[New NCERT Class 11th Page No. 77]**Q107 Text Solution:**

(3)

The breaking of C-C bonds of complex organic molecules by oxidation cells leading to the release of a lot of energy is called cellular respiration. Glucose is the favoured substrate for respiration. Fats and proteins can also be broken down to yield energy. The initial stage of cellular respiration takes place in the cytoplasm. Each glucose molecule is broken through a series of enzyme catalysed reactions into two molecules of pyruvic acid. This process is called glycolysis. The fate of the pyruvate depends on the availability of oxygen and the organism.

[New NCERT Class 11th Page No. 164]**Q108 Text Solution:**

(4)

One single maize root apical meristem can give rise to more than 17,500 new cells per hour, whereas cells in a watermelon may increase in size by upto 3,50,000 times. While the growth of a pollen tube is measured in terms of its length, an increase in surface area denotes the growth in a dorsiventral leaf.

[New NCERT Class 11th Page No. 168]**Q109 Text Solution:**

(3)

Cornelius van Neil. (1897-1985), who, based on his studies of purple and green bacteria, demonstrated that photosynthesis is essentially a light-dependent reaction in which hydrogen from

a suitable oxidisable compound reduces carbon dioxide to carbohydrates.

[New NCERT Class 11th Page No. 135]**Q110 Text Solution:**

(2)

Testa	Outer layer of seed
Tegmen	Inner layer of seed
Micropyle	Above the hilum
Aleurone	Outer covering of endosperm

[New NCERT Class 11th Page No. 66, 67]**Q111 Text Solution:**

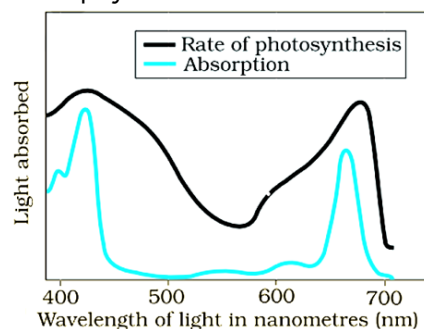
(4)

Parenchymatous cells are usually present in cortex, pericycle, pith and medullary rays, in the primary stems and roots. Hypodermis is collenchymatous in dicot stem and sclerenchymatous in monocot stem.

[New NCERT Class 11th Page No. 72]**Q112 Text Solution:**

(3)

Graph showing action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll a.

[New NCERT Class 11th Page No. 135]**Q113 Text Solution:**[Android App](#)[iOS App](#)[PW Website](#)

(1)

The ground tissue system forms the main bulk of the plant. It is divided into three zones – cortex, pericycle and pith. The vascular tissue system is formed by the xylem and phloem.

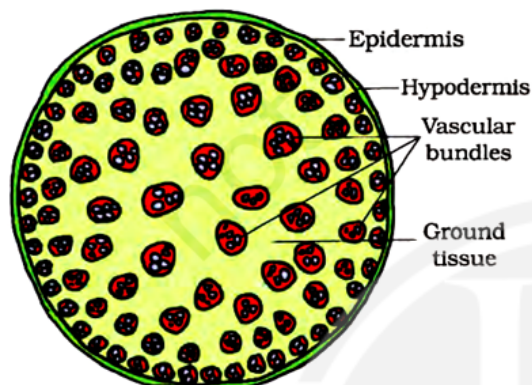
[New NCERT Class 11th Page No. 77]

Q114 Text Solution:

(2)

I-Vascular bundle

II-Ground tissue



[New NCERT Class 11th Page No. 75]

Q115 Text Solution:

(4)

A sterile stamen is called staminode.

[New NCERT Class 11th Page No. 64]

Q116 Text Solution:

(1)

In geometrical growth the initial growth is slow (lag phase), and it increases rapidly thereafter – at an exponential rate (log or exponential phase).

[New NCERT Class 11th Page No. 170]

Q117 Text Solution:

(4)

The epidermis which covers both the upper surface (adaxial epidermis) and lower surface (abaxial epidermis) of the leaf has a conspicuous cuticle. The abaxial epidermis generally bears more stomata than the adaxial epidermis. The tissue between the upper and the lower epidermis is called the mesophyll. Mesophyll, which possesses chloroplasts and carry out photosynthesis, is made up of parenchyma.

[New NCERT Class 11th Page No. 76]

Q118 Text Solution:

(1)

Ethylene enhances the respiration rate during ripening of the fruits.

[New NCERT Class 11th Page No. 177]

Q119 Text Solution:

(3)

Cytochrome c is a small protein attached to the outer surface of the inner membrane and acts as a mobile carrier for transfer of electrons between complex III and IV.

Q120 Text Solution:

(1)

In pea and bean flowers, there are five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel); this type of aestivation is known as vexillary or papilionaceous.

[New NCERT Class 11th Page No. 62, 64]

Q121 Text Solution:

(4)

Found in dicot stem only.

[New NCERT Class 11th Page No. 75, 76]

Q122 Text Solution:

(4)

Inhibitor-B is abscisic acid.

Auxin help to initiate rooting in stem cuttings.

[New NCERT Class 11th Page No. 175,176,177]

Q123 Text Solution:

(1)

Auxins like IAA and indole butyric acid (IBA) have been isolated from plants. NAA (naphthalene acetic acid) and 2, 4-D (2, 4-dichlorophenoxyacetic) are synthetic auxins.

[New NCERT Class 11th Page No. 175]

Q124 Text Solution:

(4)

In Krebs cycle there are three points where NAD^+ is reduced to $\text{NADH} + \text{H}^+$ and one point where FAD^+ is reduced to FADH_2 .

[New NCERT Class 11th Page No. 159]

Q125 Text Solution:



(1)

Generally, the fruit consists of a wall or pericarp and seeds. The pericarp may be dry or fleshy. When pericarp is thick and fleshy, it is differentiated into the outer epicarp, the middle mesocarp and the inner endocarp. In mango and coconut, the fruit is known as a drupe. They develop from monocarpellary superior ovaries and are one seeded.

[New NCERT Class 11th Page No. 65]**Q126 Text Solution:**

(4)

In dicotyledonous stems, cambium is present between phloem and xylem. Such vascular bundles because of the presence of cambium possess the ability to form secondary xylem and phloem tissues, and hence are called open vascular bundles. In the monocotyledons, the vascular bundles have no cambium present in them. Hence, since they do not form secondary tissues they are referred to as closed.

[New NCERT Class 11th Page No. 73]**Q127 Text Solution:**

(4)

NADPH + H ⁺ formation	Non cyclic photophosphorylation
ATP formation only	Stromal lamella
RuBisCO is utilised	Calvin cycle
PEPcase is utilised	OAA formation

[New NCERT Class 12th Page No. 140, 146]**Q128 Text Solution:**

(1)

- Presence of stomata
- Presence of lenticels
- Loose packing of parenchyma cells in leaves
- Each plant part takes care of its own gas-exchange needs.

[New NCERT Class 11th Page No. 154,155]**Q129 Text Solution:**

(2)

Carboxylation is the most crucial step of the Calvin cycle where CO₂ is utilised.

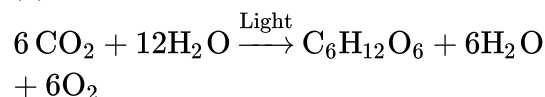
[New NCERT Class 11th Page No. 143, 144]**Q130 Text Solution:**

(2)

Rachis is present in neem.

[New NCERT Class 11th Page No. 61]**Q131 Text Solution:**

(4)



oxaloacetic acid is formed in C₄ pathway.

[New NCERT Class 11th Page No. 135]**Q132 Text Solution:**

(4)

The PGRs that are involved in growth inhibiting activities group play an important role in plant responses to wounds and stresses of biotic and abiotic origin. They are also involved in various growth inhibiting activities such as dormancy and abscission. The PGR abscisic acid belongs to this group.

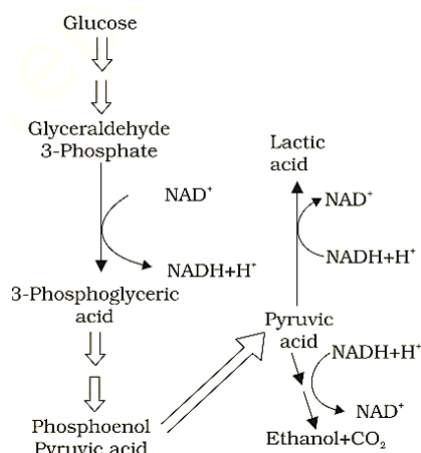
[New NCERT Class 11th Page No. 174]**Q133 Text Solution:**

(2)

The trichomes are present in the shoot system.

[New NCERT Class 11th Page No. 72]**Q134 Text Solution:**

(3)

[New NCERT Class 11th Page No. 157]**Q135 Text Solution:**

Android App

iOS App

PW Website

(4)

In symmetry, the flower may be actinomorphic (radial symmetry) or zygomorphic (bilateral symmetry). When a flower can be divided into two equal radial halves in any radial plane passing through the centre, it is said to be actinomorphic, e.g., mustard, *datura*, chilli. When it can be divided into two similar halves only in one particular vertical plane, it is zygomorphic, e.g., pea, gulmohur, bean, *Cassia*. A flower is asymmetric (irregular) if it cannot be divided into two similar halves by any vertical plane passing through the centre, as in canna.

[New NCERT Class 11th Page No. 62]**Q136 Text Solution:**

(4)

As compared to the dicot root which have fewer xylem bundles, there are usually more than six (polyarch) xylem bundles in the monocot root.

[New NCERT Class 11th Page No. 74]**Q137 Text Solution:**

(2)

Splitting of the water molecule takes place on the inner side of the thylakoid membrane, the protons or hydrogen ions that are produced by the splitting of water accumulate within the lumen of the thylakoids.

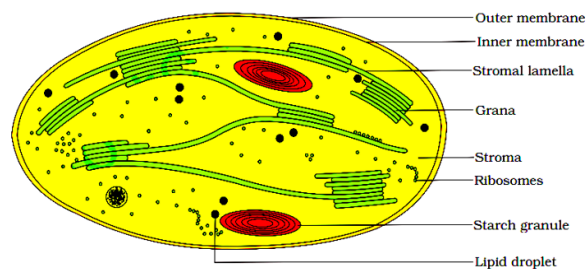
[New NCERT Class 11th Page No. 140]**Q138 Text Solution:**

(3)

H.H. Cousins	Ripened oranges
F. Skoog	Tobacco stem
E. Kurosawa	Rice seedlings
F.W. Went	Coleoptiles of oat seedlings.

[New NCERT Class 11th Page No. 175]**Q139 Text Solution:**

(4)



A - Stroma lamella

B - Grana

[New NCERT Class 11th Page No. 1362]**Q140 Text Solution:**

(3)

Mesophyll not differentiated into palisade and spongy tissues.

[New NCERT Class 11th Page No. 77]**Q141 Text Solution:**

(3)

Although the aerobic process of respiration takes place only in the presence of oxygen, the role of oxygen is limited to the terminal stage of the process. Yet, the presence of oxygen is vital, since it drives the whole process by removing hydrogen from the system. Oxygen acts as the final hydrogen acceptor.

[New NCERT Class 11th Page No. 160]**Q142 Text Solution:**

(1)

Cytokinin helps to produce new leaves, chloroplasts in leaves, lateral shoot growth and adventitious shoot formation. Cytokinins help overcome the apical dominance. They promote nutrient mobilisation which helps in the delay of leaf senescence.

[New NCERT Class 11th Page No. 177]**Q143 Text Solution:**

(1)

All tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith constitute the stele.

[New NCERT Class 11th Page No. 74]**Q144 Text Solution:**

(2)



- In dorsiventral leaf spongy parenchyma is situated below the palisade cells.
- Spongy parenchyma cells are loosely arranged.

[New NCERT Class 11th Page No. 77]

Q145 Text Solution:

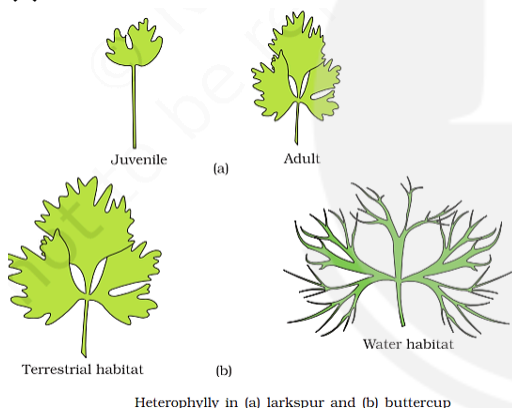
(2)

A flower is a modified shoot wherein the shoot apical meristem changes to floral meristem. Internodes do not elongate and the axis gets condensed. The apex produces different kinds of floral appendages laterally at successive nodes instead of leaves. When a shoot tip transforms into a flower, it is always solitary. The arrangement of flowers on the floral axis is termed as inflorescence.

[New NCERT Class 11th Page No. 61]

Q146 Text Solution:

(4)



Heterophylly in (a) larkspur and (b) buttercup

[New NCERT Class 11th Page No. 173]

Q147 Text Solution:

(3)

- ATP is synthesised during the conversion of phosphoenol pyruvate to pyruvic acid in glycolysis.
- Glycolysis is often referred to as the EMP pathway.

[New NCERT Class 11th Page No. 155]

Q148 Text Solution:

(2)

The pith is small or inconspicuous.

[New NCERT Class 11th Page No. 74]

Q149 Text Solution:

(4)

The formation of acetyl coenzyme A from pyruvic acid produces (in addition to acetyl coenzyme A) one molecule of CO_2 one molecule of NADH and one molecule (ion) of H^+ .

[New NCERT Class 11th Page No. 158]

Q150 Text Solution:

(4)

The root is covered at the apex by a thimble-like structure called the root cap. It protects the tender apex of the root as it makes its way through the soil. A few millimetres above the root cap is the region of meristematic activity. The cells of this region are very small, thin-walled and with dense protoplasm. They divide repeatedly. The cells proximal to this region undergo rapid elongation and enlargement and are responsible for the growth of the root in length.

[New NCERT Class 11th Page No. 59]

Q151 Text Solution:

(3)

Plasma without the clotting factors is called serum.

[New NCERT Class 11th Page No. 194]

Q152 Text Solution:

(4)

- Reptiles, birds, land snails and insects excrete nitrogenous wastes as uric acid in the form of pellet or paste with a minimum loss of water and are called uricotelic animals.
- Aquatic insects are ammonotelic in nature.

[New NCERT Class 11th Page No. 205]

Q153 Text Solution:

(1)

- Rh antigens of the foetus do not get exposed to the Rh^{ve} blood of the mother in the first pregnancy as the two bloods are well separated by the placenta.
- A clot or coagulum is formed mainly of a network of threads called fibrins in which



dead and damaged formed elements of blood are trapped.

[New NCERT Class 11th Page No. 196]

Q154 Text Solution:

(4)

Kidneys are situated between the levels of the last thoracic and third lumbar vertebra.

[New NCERT Class 11th Page No. 206]

Q155 Text Solution:

(3)

Angiotensin II, being a powerful vasoconstrictor, increases the glomerular blood pressure and thereby GFR.

[New NCERT Class 11th Page No. 212]

Q156 Text Solution:

(1)

Aldosterone causes reabsorption of Na⁺ and water from the distal parts of the tubule. This also leads to an increase in blood pressure and GFR.

[New NCERT Class 11th Page No. 212]

Q157 Text Solution:

(3)

- Nearly 70-80 percent of electrolytes and water are reabsorbed by the PCT.
- The ascending limb is impermeable to water but allows transport of electrolytes actively or passively.
- Conditional reabsorption of Na⁺ and water takes place through the DCT.

[New NCERT Class 11th Page No. 279]

Q158 Text Solution:

(2)

Mammals have the ability to produce concentrated urine. The Henle's loop and *vasa recta* play a significant role in this.

[New NCERT Class 11th Page No. 211]

Q159 Text Solution:

(4)

All vertebrates (Fishes, reptiles, amphibians and birds) have a closed circulatory system.

[New NCERT Class 11th Page No. 203]

Q160 Text Solution:

(4)

- A special neural centre in the medulla oblongata can moderate the cardiac function through the autonomic nervous system.
- Parasympathetic neural signals can decrease cardiac output.

[New NCERT Class 11th Page No. 202]

Q161 Text Solution:

(1)

Each kidney has nearly one million complex tubular structures called nephrons.

[New NCERT Class 11th Page No. 207]

Q162 Text Solution:

(2)

- Unipolar neurons are found in the embryonic stages.
- Bipolar neurons have a cell body with one axon and one dendrite.

[New NCERT Class 11th Page No. 232]

Q163 Text Solution:

(3)

- The second heart sound (dub) is associated with the closure of the semilunar valves
- whereas the first heart sound (lub) is associated with the closure of the tricuspid and bicuspid valves.
- Human heart is the mesodermally derived organ, situated in the thoracic cavity.

[New NCERT Class 11th Page No. 200]

Q164 Text Solution:

(4)

- ADH (Antidiuretic hormone) :
Neurohypophysis Renin : Kidney

[New NCERT Class 11th Page No. 212]

Q165 Text Solution:

(3)



Two separate circulatory pathways are present in the heart of birds and mammals and hence, these animals have double circulation.

[New NCERT Class 11th Page No. 197]

Q166 Text Solution:
(3)

- Phalanges are digits and 14 in number in each arms and legs.
- Carpals are wrist bones that are 8 in number in each arm.

[New NCERT Class 11th Page No. 226]

Q167 Text Solution:
(3)

The skull is composed of two sets of bones i.e., cranial and facial bones.

[New NCERT Class 11th Page No. 224]

Q168 Text Solution:
(1)

Myasthenia gravis is an auto immune disorder affecting neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

[New NCERT Class 11th Page No. 227]

Q169 Text Solution:
(3)

The vertebral column is differentiated into cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused) regions starting from the skull.

[New NCERT Class 11th Page No. 226]

Q170 Text Solution:
(1)

- Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called uremia, which is highly harmful and may lead to kidney failure. In such patients, urea can be removed by a process called hemodialysis.
- Kidney transplantation is the ultimate method in the correction of acute renal failures.

[New NCERT Class 11th Page No. 213]

Q171 Text Solution:
(2)

Each rib is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum.

[New NCERT Class 11th Page No. 218, 219, 225]

Q172 Text Solution:
(1)

- Fibrous joints do not allow any movement.
- Fibrous joint is shown by the flat skull bones which fuse end-to-end with the help of dense fibrous connective tissues in the form of sutures, to form the cranium.

[New NCERT Class 11th Page No. 227]

Q173 Text Solution:
(4)

Renal calculi	Stones formed within the kidney
Glomerulonephritis	Inflammation of glomeruli of kidney
Arthritis	Inflammation of joints
Muscular dystrophy	Genetic disorder

[New NCERT Class 11th Page No. 214, 227]

Q174 Text Solution:
(4)

- Scapula is situated in the dorsal part of the thorax between the second and the seventh ribs of the appendicular skeleton.
- The central part of thick filament, not overlapped by thin filaments is called the 'H' zone.

[New NCERT Class 11th Page No. 220, 226]

Q175 Text Solution:
(2)

In human beings, ammonia produced by metabolism is converted into urea in the liver and then excreted out by kidneys.



[New NCERT Class 11th Page No. 205]**Q176 Text Solution:**

(3)

Substances like glucose, amino acids, Na^+ , etc., in the filtrate are reabsorbed actively whereas the nitrogenous wastes are absorbed by passive transport.

[New NCERT Class 11th Page No. 209]**Q177 Text Solution:**

(3)

The osmotic concentration of glomerular filtrate in the cortex region of kidneys is about 300 mOsmL^{-1} .

[New NCERT Class 11th Page No. 211]**Q178 Text Solution:**

(1)

- Terrestrial adaptation necessitated the production of lesser toxic nitrogenous wastes like urea and uric acid for conservation of water.
- Nearly all of the essential nutrients, and 70-80 per cent of electrolytes and water are reabsorbed by the proximal convoluted tubule.

[New NCERT Class 11th Page No. 205, 209]**Q179 Text Solution:**

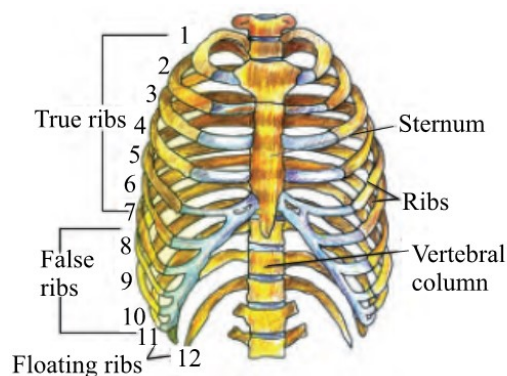
(3)

Each sarcomere has a central 'A' band made of thick myosin filaments, and two half 'I' bands made of thin actin filaments.

[New NCERT Class 11th Page No. 220]**Q180 Text Solution:**

(3)

The labeling "A" represents the vertebral column.

[New NCERT Class 11th Page No. 225, 226]**Q181 Text Solution:**

(3)

Cerebrum	Seat of intelligence
Cerebellum	Maintains body posture
Hypothalamus	Controls the pituitary
Midbrain	Controls vision and hearing

[New NCERT Class 11th Page No. 236]**Q182 Text Solution:**

(3)

Sympathetic nervous system is activated releasing epinephrine and norepinephrine from adrenal medulla.

[New NCERT Class 11th Page No. 231]**Q183 Text Solution:**

(1)

Iodine is essential for the normal rate of hormone synthesis in the thyroid. Deficiency of iodine in our diet results in hypothyroidism and enlargement of the thyroid gland, commonly called goitre. Hypothyroidism during pregnancy causes defective development and maturation of the growing baby leading to stunted growth (cretinism), mental retardation, low intelligence quotient, abnormal skin, deaf-mutism, etc.

[New NCERT Class 11th Page No. 334]**Q184 Text Solution:**

(4)

- Along with the hypothalamus, the limbic system is involved in the regulation of sexual



behaviour, expression of emotional reactions (e.g., excitement, pleasure, rage and fear), and motivation.

- Pia mater is innermost layer of meninges.
- The midbrain receives and integrates visual, tactile and auditory inputs.
- A deep cleft divides the cerebrum longitudinally into two halves, which are termed as the left and right cerebral hemispheres.
- The human brain is well protected by the bony skull.

[New NCERT Class 11th Page No. 236, 237]

Q185 Text Solution:

(4)

Melatonin plays a very important role in the regulation of a 24-hour (diurnal) rhythm of our body.

[New NCERT Class 11th Page No. 242]

Q186 Text Solution:

(3)

- Red skeletal muscle fibres contain large amounts of myoglobin which imparts reddish appearance to them.
- They have less sarcoplasmic reticulum.
- These muscles contain plenty of mitochondria which can utilise the large amount of oxygen stored in them for ATP production. These muscles, therefore, can also be called aerobic muscles.

[New NCERT Class 11th Page No. 222]

Q187 Text Solution:

(1)

- In polarised axonal membrane, the outer surface possesses a positive charge while its inner surface becomes negatively charged and therefore is polarised.
- Sodium-potassium pump actively transports 3 Na⁺ outwards for 2K⁺ into the cell.

[New NCERT Class 11th Page No. 232]

Q188 Text Solution:

(3)

The two cerebral hemispheres are connected by a tract of nerve fibres called corpus callosum.

[New NCERT Class 11th Page No. 236]

Q189 Text Solution:

(1)

In adrenal gland, centrally located tissue is called the adrenal medulla, and outside this lies the adrenal cortex.

[New NCERT Class 11th Page No. 243]

Q190 Text Solution:

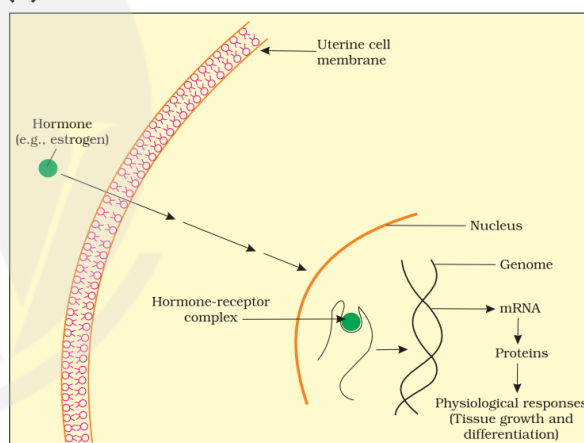
(2)

In Addison's disease, underproduction of hormones by the adrenal cortex alters carbohydrate metabolism causing acute weakness and fatigue.

[New NCERT Class 11th Page No. 243]

Q191 Text Solution:

(2)



[New NCERT Class 11th Page No. 341]

Q192 Text Solution:

(2)

- Angina pectoris occurs when enough oxygen is not reaching the heart muscles. A symptom of acute chest pain occurs in it.
- Angina can occur in men and women of any age but it is more common among the middle-aged and elderly.
- It occurs due to conditions that affect the blood flow.

[New NCERT Class 11th Page No. 101]



Q193 Text Solution:**(4)**

- Ovary is the primary female sex organ that produces one ovum during each menstrual cycle.
- In human female ovary, estrogen is synthesised and secreted mainly by the growing ovarian follicles.

[New NCERT Class 11th Page No. 246]**Q194 Text Solution:****(2)**

The CNS (central nervous system) includes the brain and the spinal cord.

[New NCERT Class 11th Page No. 231]**Q195 Text Solution:****(4)**

Pineal gland	Melatonin
Thyroid gland	Thyroxine
Ovary	Estrogen
Adrenal medulla	Epinephrine

[New NCERT Class 11th Page No. 242, 243, 244, 245, 246]**Q196 Text Solution:****(1)**

Platelets, also called thrombocytes, are cell fragments produced from megakaryocytes (special cells in the bone marrow).

[New NCERT Class 11th Page No. 195]**Q197 Text Solution:****(4)**

- Antennal glands or green glands perform the excretory function in crustaceans like prawns.
- Malpighian tubules perform the excretory function in most of the insects including the cockroaches.

[New NCERT Class 11th Page No. 206]**Q198 Text Solution:****(2)**

- The 8th, 9th and 10th pairs of ribs do not articulate directly with the sternum but join the seventh rib with the help of hyaline cartilage. These are called vertebrochondral or false ribs.
- Last 2 pairs (11th and 12th) of ribs are not connected ventrally and are therefore, called floating ribs.

[New NCERT Class 11th Page No. 225]**Q199 Text Solution:****(1)**

- Plasma is a straw-coloured, viscous fluid constituting nearly 55% of the blood.
- Fats are absorbed through lymph in the lacteals present in the intestinal villi.
- In cardiac arrest, heart stops beating. Globulins primarily are involved in defense mechanisms of the body.
- Mitral valve is present between left auricle and left ventricle.

[New NCERT Class 11th Page No. 201]**Q200 Text Solution:****(3)**

- Ascending limb of loop of Henle is impermeable to water and allow transport of electrolyte actively and passively.
- As the concentrated filtrate pass upward, it gets diluted due to the passage of electrolytes to the medullary fluid.

[New NCERT Class 11th Page No. 244]

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